

Notes: Rosenberg & Engle (JFE, 2002)

Empirical Pricing Kernels

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1 Big picture

- **Question.** Can we recover a *time-varying* pricing kernel from option prices once we pair those prices with an estimated objective distribution of future stock-market returns? And if we do so, does the estimated kernel reveal economically meaningful variation in risk aversion?
- **Setting.** The paper uses one-month S&P 500 index option prices and an estimated S&P 500 return density to construct a monthly empirical pricing kernel (EPK) from 1991 to 1995.
- **Core challenge.** Option prices identify *state prices*, not the pricing kernel alone. To back out the kernel, the paper must separately estimate the objective probability density of future returns. A poor density estimate mechanically contaminates the kernel estimate.
- **Main conceptual contribution.** The paper defines the **empirical pricing kernel** as the state-dependent preference function that gives the *best fit* to observed option prices, conditional on a forecast return density. This makes the kernel a date-by-date object rather than an average object over a long historical window.
- **Why this differs from earlier work.** Earlier option-based papers often estimated a pricing kernel by smoothing realized returns over multi-year windows. Rosenberg and Engle instead estimate a *dynamic* return density with an asymmetric GARCH model and re-estimate the kernel each month, so both probabilities and risk aversion can move over time.
- **Main empirical design.** Each month, the authors:
 - estimate the S&P 500 return state density using an asymmetric GARCH model plus an empirical innovation density;
 - choose pricing-kernel parameters to minimize the distance between fitted and actual option prices;
 - compare a **power** kernel with a more flexible **orthogonal-polynomial** kernel;
 - relate the implied risk-aversion parameter to business-cycle indicators; and
 - evaluate the models using **out-of-sample hedging performance**.
- **Main pricing result.** The pricing kernel is clearly *downward sloping* over S&P 500 return states, so a unit payoff is more valuable in bad market states than in good states.
- **Main time-variation result.** Under the power specification, the projected risk-aversion coefficient averages **7.36**, ranges from **2.36** to **12.55**, and is positively autocorrelated with first-order autocorrelation about **0.45**.
- **Main functional-form result.** The orthogonal-polynomial kernel fits option prices more closely, with average pricing-error standard deviation about **\$0.09**, versus roughly **\$0.63** for the power kernel.
- **Main macro-finance result.** Estimated risk aversion is **countercyclical**: it rises when credit spreads widen and falls when the term structure steepens.
- **Main validation result.** Time-varying pricing kernels improve hedging performance relative to time-invariant kernels by roughly **1% to 3%** in hedge-portfolio volatility, and the **power** kernel hedges better than the more flexible orthogonal-polynomial kernel.

- **Economic takeaway.** Option prices contain useful information about changing state prices and changing risk aversion. But the specification that fits prices best is not automatically the one that hedges best, so both *pricing fit* and *economic performance* matter.

2 Data and measurement (key objects)

2.1 Why S&P 500 index options are the right laboratory

- Options have state-contingent payoffs, so their prices are especially informative about how investors value different return states.
- Using options avoids the need to infer the pricing kernel from aggregate consumption data, which the paper argues is often noisy and difficult to measure accurately.
- The paper projects the kernel onto **S&P 500 return states**. This is natural because the option payoffs themselves are functions of the future S&P 500 level.

Interpretation. The option market gives a direct cross-section of prices for payoffs that load differently on good and bad equity-market states.

2.2 Option data source and construction of closing prices

- The options data come from the **Berkeley Options Database**.
- Sample period for kernel estimation: **1991–1995**.
- For each contract, the paper forms an end-of-day quote by averaging the last bid-ask quote recorded between **2:00 and 3:00 PM Central Time**.
- Because quotes in the cross-section are not perfectly synchronized, the authors first back out a Black–Scholes implied volatility for each option and then re-price the option at the *closing* S&P 500 index level.
- They then average each call price with its synthetic put counterpart, and each put price with its synthetic call counterpart, using put–call parity adjusted for dividends.

Interpretation. Black–Scholes is used here mainly as an *interpolation / synchronization device*, not as the maintained structural pricing model.

2.3 Screening rules for usable option observations

The paper keeps only options that satisfy a fairly strict set of filters:

- moneyness between -10% and $+10\%$, measured by $(K/S_t - 1)$;
- midquote between **\$3/8** and **\$50**;
- annualized implied volatility between **5%** and **90%**;

- no-arbitrage lower-bound conditions;
- monotonicity of option premia in strike; and
- vertical-spread premium restrictions.

The estimation sample uses only dates with at least **eight options** passing the screens.

Interpretation. The paper wants a cross-section that is both economically informative and clean enough that pricing-kernel estimates are not driven by stale quotes or obvious data errors.

2.4 Monthly one-month option sample

- The authors target options with about **20 trading days** to expiration.
- They first discard options with fewer than **16** or more than **24** trading days to expiry.
- On each trading date, they keep the series with maturity closest to 20 days.
- Out of 60 possible months, they obtain **53 monthly estimation dates**.
- On a given date, the usable cross-section contains between **8 and 13** options.

Interpretation. The kernel is estimated from a standardized one-month horizon rather than mixing widely different option maturities.

2.5 Why only out-of-the-money calls and puts enter the estimation

- By put–call parity, a call and a put with the same strike contain the same information once the underlying asset and interest rates are accounted for.
- So the estimation uses only **out-of-the-money put options** (negative moneyness) and **out-of-the-money call options** (positive moneyness).

Interpretation. This avoids redundant observations without discarding economically relevant strike variation.

2.6 Underlying return data for the objective density

- To estimate the S&P 500 return process, the paper uses daily S&P 500 log returns from **1970–1995**.
- The series contains **6,571 observations**.
- Summary statistics:
 - annualized mean return: **7.55%**;
 - annualized standard deviation: **14.79%**;
 - skewness: **−2.31**;

– excess kurtosis: **60.20**.

Interpretation. The raw return series is strongly non-normal and exhibits volatility clustering, so any plausible objective density must allow heavy tails and stochastic volatility.

2.7 Risk-free rates, dividends, and other pricing inputs

- Risk-free rates come from Datastream bid and ask discount rates on one-, three-, and six-month Treasury bills.
- For each option, the relevant risk-free rate is linearly interpolated to the contract’s expiration.
- Dividend yields are constructed from the present value of future S&P 500 dividends divided by the current index level.

Interpretation. These inputs are needed only to convert raw option quotes into internally consistent end-of-day option prices and implied volatilities.

2.8 The key latent object: the projected pricing kernel

Let the full pricing kernel be $M_t(Z_t, Z_{t+1})$, where Z_{t+1} contains all payoff-relevant future state variables. The paper focuses on the pricing kernel projected onto the traded asset payoff state:

$$M_t^*(X_{t+1}) = E_t[M_t(Z_t, Z_{t+1}) \mid X_{t+1}]. \quad (1)$$

Interpretation. This is the part of the full kernel that matters for pricing assets whose payoffs depend only on X_{t+1} , here the future S&P 500 return or index level.

2.9 Projected risk aversion

The paper defines projected risk aversion as

$$g_t^* = -\frac{X_{t+1}M_t^{*'}(X_{t+1})}{M_t^*(X_{t+1})}. \quad (2)$$

Interpretation. A larger value means the projected kernel is more steeply downward sloping, so investors place a higher relative value on payoffs delivered in bad states.

2.10 Business-cycle indicators used later

To link estimated risk aversion to macro conditions, the paper studies:

- the **credit spread**: Moody’s long-term Baa corporate yield minus the 30-year Treasury yield;
- the **term spread**: 30-year Treasury yield minus three-month Treasury yield;

- the change in the **three-month Treasury yield**;
- the one-month **S&P 500 return**;
- **consumption growth**; and
- the spread between **implied** and **objective** volatility.

Interpretation. These variables proxy for recessionary vs expansionary conditions and allow the authors to test whether risk aversion is countercyclical.

3 Models and empirical specifications (all equations + interpretation)

3.1 Basic no-arbitrage asset-pricing equation

The starting point is the standard pricing-kernel formula:

$$P_t = E_t[M_t X_{t+1}], \quad (3)$$

where P_t is today's asset price, X_{t+1} is next period's payoff, and M_t is the stochastic discount factor.

Interpretation. Asset prices are expected payoffs weighted by how much investors value cash in each state.

3.2 Projected pricing kernel for assets whose payoff depends on one state variable

If the payoff depends only on X_{t+1} , the pricing equation can be rewritten as

$$P_t = E_t[M_t^*(X_{t+1})X_{t+1}], \quad M_t^*(X_{t+1}) = E_t[M_t(Z_t, Z_{t+1}) \mid X_{t+1}]. \quad (4)$$

Interpretation. The full multidimensional kernel is replaced by a one-dimensional object indexed by future S&P return states. This is what makes estimation from options feasible.

3.3 Derivative pricing equation used in estimation

For derivative i with payoff $g_i(r_{t+1})$, price $P_{i,t}$, and objective return density $f_t(r_{t+1})$, the paper writes

$$P_{i,t} = E_t[M_t^*(r_{t+1})g_i(r_{t+1})] = \int M_t^*(r_{t+1})g_i(r_{t+1})f_t(r_{t+1})dr_{t+1}. \quad (5)$$

Interpretation. Option prices reveal state-price weights. Once the paper specifies $f_t(\cdot)$, the remaining unknown object is the pricing kernel itself.

3.4 Definition of the empirical pricing kernel (EPK)

Let $M_t^*(r_{t+1}; y_t)$ be a parametric pricing-kernel candidate indexed by parameter vector y_t . The fitted option price is

$$\hat{P}_{i,t}(y_t) = \int M_t^*(r_{t+1}; y_t) g_i(r_{t+1}) \hat{f}_t(r_{t+1}) dr_{t+1}. \quad (6)$$

The empirical pricing kernel solves

$$\min_{y_t} \sum_{i=1}^L \left(P_{i,t} - \hat{P}_{i,t}(y_t) \right)^2. \quad (7)$$

Interpretation. The EPK is simply the kernel that *best reproduces the observed option cross-section* on that date, given the estimated objective density.

3.5 Simulation approximation

With J simulated future return realizations $\{r_{t+1,j}\}_{j=1}^J$, the fitted price is approximated by

$$\hat{P}_{i,t}(y_t) \approx \frac{1}{J} \sum_{j=1}^J M_t^*(r_{t+1,j}; y_t) g_i(r_{t+1,j}). \quad (8)$$

The paper uses **200,000 Monte Carlo replications** in the empirical implementation.

Interpretation. The integral is turned into an average over a simulated return distribution generated by the estimated GARCH model.

3.6 Power pricing-kernel specification

The first specification is

$$M_t^*(r_{t+1}; y_t) = y_{0,t} (r_{t+1})^{-y_{1,t}}. \quad (9)$$

In this case, projected risk aversion is simply

$$g_t^* = y_{1,t}. \quad (10)$$

Interpretation. The single slope parameter $y_{1,t}$ is the paper's cleanest empirical measure of risk aversion. When $y_{1,t}$ rises, bad states are priced more aggressively.

3.7 Orthogonal-polynomial pricing-kernel specification

The flexible alternative is a generalized Chebyshev expansion:

$$M_t^*(r_{t+1}; y_t) = [y_{0,t} T_0(r_{t+1})] \exp(y_{1,t} T_1(r_{t+1}) + y_{2,t} T_2(r_{t+1}) + \cdots + y_{N,t} T_N(r_{t+1})). \quad (11)$$

Empirically, the paper uses a **four-parameter** version with terms through T_3 .

Interpretation. This specification can bend, twist, and even create regions of locally increasing marginal utility, so it is much more flexible than the power form.

3.8 Bond-pricing normalization

To ensure that the kernel prices a one-month riskless bond correctly, the scale factor $y_{0,t}$ is set to satisfy

$$B_t = E_t[M_t^*(r_{t+1}; y_t)]. \quad (12)$$

Interpretation. The cross-section of options identifies the *shape* of the kernel, but the bond-pricing restriction pins down its overall level.

3.9 Objective S&P 500 return density: asymmetric GARCH

The log-return equation is

$$\ln(S_t/S_{t-1}) - r_f = \mu + e_t, \quad e_t \sim f(0, \sigma_{t|t-1}^2), \quad (13)$$

with conditional variance

$$\sigma_{t|t-1}^2 = \omega_1 + \omega_2 I + \alpha e_{t-1}^2 + \beta \sigma_{t-1|t-2}^2 + \delta \max(0, -e_{t-1})^2. \quad (14)$$

Interpretation.

- αe_{t-1}^2 captures ARCH effects;
- $\beta \sigma_{t-1|t-2}^2$ captures persistent volatility;
- $\delta \max(0, -e_{t-1})^2$ allows **negative returns** to increase volatility more than positive returns of the same size.

3.10 Empirical innovation density

Instead of forcing the innovations to be Gaussian, the paper constructs an empirical innovation density from standardized residuals:

$$\frac{e_t}{\sigma_{t|t-1}}. \quad (15)$$

Interpretation. The GARCH part handles *time-varying scale*; the empirical residual distribution handles *non-normal shape*, including skewness, excess kurtosis, and extreme crashes.

3.11 Hedge-ratio specification

For one-day changes in an option's value, the paper uses a second-order Taylor approximation:

$$\text{Put}_{t+1} - \text{Put}_t \approx \frac{\partial \text{Put}_{t+1}}{\partial S_{t+1}}(S_{t+1} - S_t) + \frac{1}{2} \frac{\partial^2 \text{Put}_{t+1}}{\partial S_{t+1}^2}(S_{t+1} - S_t)^2. \quad (16)$$

It then computes delta and gamma by centered finite differences:

$$\frac{\partial \text{Put}_{t+1}}{\partial S_{t+1}} \approx \frac{\text{Put}_{t+1|S_t+\varepsilon} - \text{Put}_{t+1|S_t-\varepsilon}}{2\varepsilon}, \quad (17)$$

$$\frac{\partial^2 \text{Put}_{t+1}}{\partial S_{t+1}^2} \approx \frac{\text{Put}_{t+1|S_t+\varepsilon} - 2\text{Put}_{t+1|S_t} + \text{Put}_{t+1|S_t-\varepsilon}}{\varepsilon^2}. \quad (18)$$

Interpretation. The hedging exercise is a practical test: a better pricing kernel should generate better deltas and gammas.

3.12 Dynamic forecasting model for next day's kernel parameters

For hedging, the time-varying kernel parameters are forecast using a simple autoregression:

$$y_{t+1} = a + b_0 y_t + \dots + b_K y_{t-K} + \varepsilon_{t+1}. \quad (19)$$

For the power specification, the most important equation is

$$y_{1,t+1} \approx 2.26 + 0.69 y_{1,t}. \quad (20)$$

Interpretation. Risk aversion is persistent but mean-reverting, which is exactly what the monthly estimates suggest.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the pricing kernel on a given date

- The option cross-section pins down state prices for S&P return states.
- The estimated objective density $\hat{f}_t(r)$ translates those state prices into a projected pricing kernel.
- The riskless-bond pricing equation fixes the overall scale.

Interpretation. Identification is not coming from a structural model of preferences; it comes from the *combination* of no-arbitrage, the option cross-section, and an assumed objective density.

4.2 Why the dynamic density model matters

The paper argues that simple historical histograms with fixed lookback windows are a weak way to infer investors' beliefs:

- they give equal weight to old and recent events inside the window;
- they abruptly drop events once they leave the window;
- they average away time variation in volatility and tail risk.

Interpretation. If the denominator of “state price per unit probability” is badly measured, the inferred pricing kernel will also be badly measured. The GARCH density is therefore a core identifying input, not a side detail.

4.3 Why the paper works with a projected kernel instead of the full kernel

- The full kernel may depend on many unobserved state variables.
- But for pricing assets whose payoffs depend only on the S&P 500 return, the projected kernel is sufficient.

Interpretation. This is a deliberate dimensionality reduction. The paper does *not* claim to recover investors' complete marginal utility function in all states of the world.

4.4 Data-quality protections

Several design choices reduce the chance that the results are driven by bad option data rather than economics:

- time-synchronization of quotes at the close;
- no-arbitrage filters;
- removal of obviously stale or implausible implied volatilities;
- use of both traded and synthetic prices via put–call parity.

Interpretation. The authors spend a lot of effort making sure the input option surface is internally consistent before any pricing-kernel estimation starts.

4.5 Why hedging is an important validation exercise

In-sample price fit alone can be misleading, especially for flexible specifications. The paper therefore asks whether a kernel that looks good in the cross-section also produces *better hedge ratios* for nearby option positions.

Interpretation. Hedging performance is a strong economic check because it tests whether the estimated kernel captures the relevant local pricing sensitivities, not just average price levels.

4.6 What the paper can and cannot claim

- It **can** estimate a projected, date-specific pricing kernel over S&P 500 return states.
- It **can** compare how different parametric kernels fit prices and hedge positions.
- It **cannot** nonparametrically identify the entire stochastic discount factor for all assets and all state variables.
- Its conclusions depend on the maintained objective-density model.

Interpretation. The results are informative but still model-dependent. The biggest maintained assumption is the return-density specification.

5 Tables and figures

5.1 Table 1 and Figure 1: the monthly option cross-sections

Table 1
Summary of option data used for pricing kernel estimation
The source of our data is the Berkeley Options Database (1991–1995). Each month, we extract from the database a cross-section of options (both puts and calls) with approximately one month until expiration. We include options with moneyness between 10% and –10%. Additional data screening criteria are used to ensure that our option closing prices are representative of market conditions at the end of the trading day. There are 53 estimation dates in the sample, and the number of observations per estimation date ranges from eight to 13.

Option moneyness ($K/S_t - 1$) (%)	Number of observations	Average implied volatility (%)	Average call price	Average put price
–6 to 10	66	20.85	\$34.83	\$1.19
–3 to 6	133	16.58	\$21.81	\$1.92
0 to 3	136	13.67	\$10.76	\$4.13
0 to 3	137	11.36	\$3.25	\$9.77
3 to 6	37	11.61	\$0.90	\$16.66

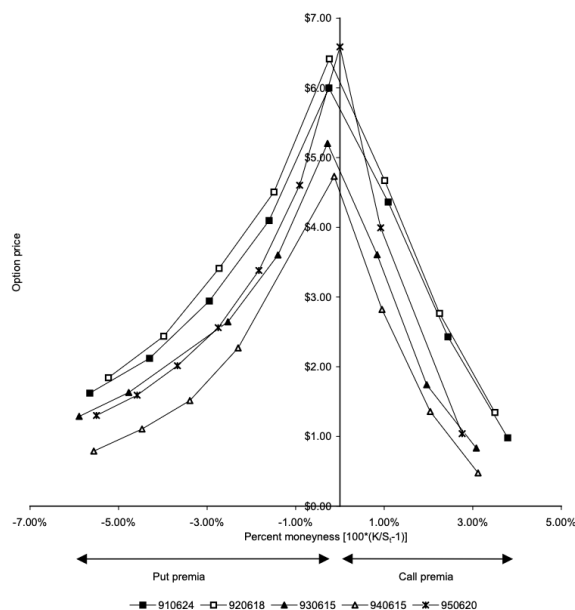


Fig. 1. One-month S&P 500 index option prices. This figure graphs the cross-section of one-month S&P 500 option prices from June 1991 to 1995 against percent moneyness (option exercise price/closing S&P 500 index level – 1) 100. For positive (negative) percent moneyness, we report call (put) premia.

Read.

- The monthly estimation sample contains **53 dates**, with between **8 and 13 options** per date.
- Most observations are concentrated in moneyness bins from about **–6% to +3%**.

- Higher-moneyness contracts have lower implied volatilities, showing the familiar **volatility skew**.
- Figure 1 shows an inverted-V shape in option premia across moneyness: put premia rise with strike while call premia fall with strike.

Economic message. The paper is fitting a fairly small but carefully filtered cross-section each month, and the cross-section clearly changes over time in ways that reflect both changing probabilities and changing state prices.

5.2 Tables 2–4 and Figures 2–3: the objective return density

Table 3
Estimated state probability models
We estimate an ARCH(1), GARCH(1,1), and asymmetric GARCH(1,1) model by maximizing the likelihood function of daily S&P 500 log-returns from 1970 to 1995. We calculate robust t-statistics using the Bollerslev and Wooldridge (1992) method. We define the asymmetric GARCH model as $\ln(S_t/S_{t-1}) - \mu = \epsilon_t$, $\epsilon_t \sim f(\theta, \sigma_{\epsilon,t}^2)$, $\sigma_{\epsilon,t}^2 = \omega_1 + \omega_2 I + \alpha_1^2 \epsilon_{t-1}^2 + \beta \sigma_{\epsilon,t-2}^2 + \delta \max(0, -\epsilon_{t-1})^2$.

The ARCH p-value is the p-value of the Engle (1982) ARCH LM statistic using ten lags. The LR (likelihood ratio) test p-value measures the statistical significance of the improvement in expanding the model specification. The reported p-value is the p-value for twice the difference between the log-likelihood of the unrestricted and restricted models.

	ARCH(1)		GARCH(1,1)		Asymmetric GARCH(1,1)	
	Coefficient	Robust t-statistic	Coefficient	Robust t-statistic	Coefficient	Robust t-statistic
μ	0.0004	3.09	0.0004	4.10	0.0003	2.41
θ	0.1483	6.71	0.1363	10.62	0.1361	10.60
ω_1	7.06E-05	17.85	1.22E-06	3.45	1.20E-06	3.80
ω_2	-3.41E-05	-8.52	-4.64E-07	-1.77	-4.34E-07	-1.69
α	0.1964	2.86	0.0663	2.41	0.0262	3.12
β			0.9212	35.88	0.9264	53.83
δ					0.0072	2.41
Log-likelihood	21,941.39		22,332.09		22,368.10	
ARCH p-value	<0.0001		0.3426		0.7004	
LR test p-value	N/A		<0.0001		<0.0001	

Table 2
S&P 500 daily log-return summary statistics, 1970–1995
The normality test p-value is the p-value of the Jarque and Bera (1980) normality test statistic, which measures the closeness of the empirical S&P 500 log-return density to a normal density. The serial correlation p-value is the p-value of the Ljung-Box Q-statistic (1978), which measures serial correlation in the innovations using ten lagged values. The ARCH test p-value is the p-value of the Engle (1982) ARCH LM statistic, which measures the presence of stochastic volatility as represented by persistence in return magnitudes. We use ten return lags for this test.

Number of observations	6571
Annualized mean	7.55%
Annualized std. dev.	14.79%
Skewness	-2.31
Excess kurtosis	60.20
Normality test p-value	<0.0001
Serial correlation test p-value	<0.0001
ARCH test p-value	<0.0001

Table 4
Properties of the standardized innovations using the estimated asymmetric GARCH model
We calculate the standardized innovations by dividing each ordinary innovation (ϵ_t) by its conditional standard deviation ($\sigma_{\epsilon,t}$). Both variables are taken from the estimated asymmetric GARCH model. In Panel A, we use Engle (1982) ARCH test statistic to measure unexplained stochastic volatility, and we use the Ljung and Box (1978) test statistic to measure unexplained serial correlation. We compare probabilities of extreme innovations for the standardized innovation density and a standard normal density in Panel B.

Panel A. Summary statistics

	Number of observations	Mean	Std. dev.	Skewness	Excess kurtosis	Normality test p-value	ARCH test p-value	Serial correlation test p-value
	6,571	0.0004	1.0000	-0.36	4.26	<.0001	0.7004	0.7378

Panel B. Probabilities of extreme events

	Empirical density	Standard normal
Probability		
Innovation < -10 std. dev.	0.00030	<0.000001
Innovation < -5 std. dev.	0.00061	<0.000001
Innovation < -3 std. dev.	0.00427	0.00135
Innovation > 3 std. dev.	0.00290	0.00135
Innovation > 5 std. dev.	0.00015	<0.000001
Innovation > 10 std. dev.	<0.000001	<0.000001

Read.

- Daily S&P returns from 1970–1995 are strongly non-normal: skewness is -2.31 and excess kurtosis is **60.20**.
- The asymmetric GARCH model dominates simpler ARCH and GARCH alternatives. The asymmetry parameter is positive and statistically significant, so bad returns raise volatility more than good returns.
- Standardized innovations from the asymmetric GARCH model pass residual serial-correlation and ARCH tests, with p-values about **0.7378** and **0.7004**.
- The empirical innovation density has skewness about -0.36 and excess kurtosis about **4.26**.
- Extreme left-tail events are much more likely under the empirical density than under a Gaussian benchmark.
- Figure 2 shows that June 1991 and June 1992 place more probability mass on large negative one-month returns than later Junes.
- Figure 3 shows annualized conditional volatility ranging from about **6.75%** to **112.86%**, with the highest values around the 1987 crash.

Economic message. The denominator of the pricing-kernel calculation is highly time-varying and strongly non-Gaussian. This is exactly why the paper refuses to rely on a simple historical histogram.

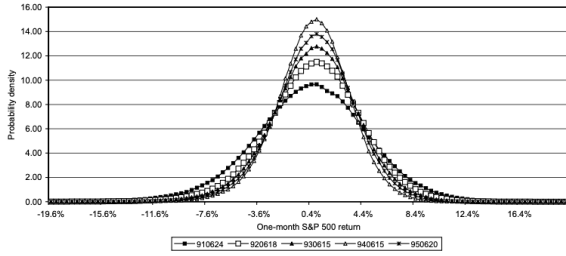


Fig. 2. Estimated S&P 500 return density functions in mid-June 1991–1995. We estimate the density using Monte Carlo simulation (200,000 replications) of the estimated asymmetric GARCH model with an empirical innovation density. We obtain density functions using a Gaussian kernel density estimator with bandwidth equal to $0.9N^{-1/5}\sigma$, where σ is the return standard deviation and N is the number of replications.

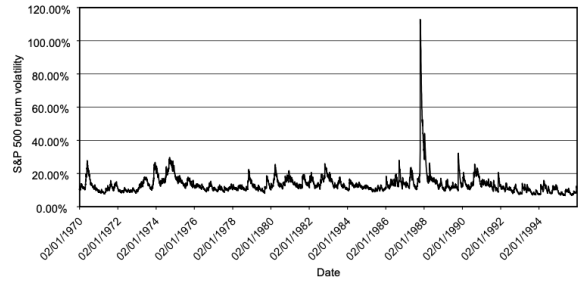


Fig. 3. Annualized S&P 500 conditional volatility, using the asymmetric GARCH model over the period 1970–1995. Dates (x-axis) follow the month/day/year format.

5.3 Table 5 and Figures 4–6: the estimated pricing kernels

Table 5

Empirical pricing kernel estimation results

We estimate one-month pricing kernels projected onto S&P 500 return states over the period 1991–1995 using two specifications. We report results for the power specification, $M_t(r_{t+1}) = \theta_{0,t}(r_{t+1})^{-\theta_{1,t}}$, in Panel A. We report results for the orthogonal polynomial specification, $M_t(r_{t+1}) = \theta_{0,t}T_0(r_{t+1}) + \theta_{2,t}T_2(r_{t+1}) + \theta_{3,t}T_3(r_{t+1})$, in Panel B.

$N = 53$	Mean	Standard deviation	Min	Max
Panel A. Power specification parameters				
$\theta_{0,t}$	1.0051	0.0063	0.9866	1.0185
$\theta_{1,t}$	7.36	2.58	2.36	12.55
Std. dev. of pricing errors	\$0.63	\$0.26	\$0.28	\$1.34
Panel B. Orthogonal polynomial specification parameters				
$\theta_{0,t}$	0.19	0.10	0.04	0.40
$\theta_{1,t}$	-2.25	1.06	-4.38	-0.25
$\theta_{2,t}$	-0.88	0.68	-2.52	0.19
$\theta_{3,t}$	-1.08	0.42	-1.94	-0.19
Std. dev. of pricing errors	\$0.09	\$0.05	\$0.03	\$0.24

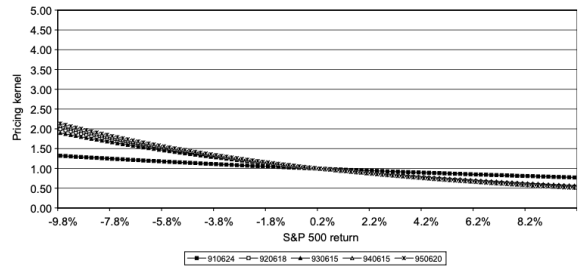


Fig. 4. Empirical pricing kernel using power specification in mid-June 1991–1995. We define this pricing kernel as $M_t(r_{t+1}) = \theta_{0,t}(r_{t+1})^{-\theta_{1,t}}$. The time-varying slope of the pricing kernel estimates reflects fluctuations in empirical risk aversion.

Read.

- Under the **power** specification, the scale factor $y_{0,t}$ averages about **1.0051** and the risk-aversion coefficient $y_{1,t}$ averages **7.36**.
- Under the **orthogonal-polynomial** specification, the average coefficients are about

$$y_{0,t} = 0.19, \quad y_{1,t} = -2.25, \quad y_{2,t} = -0.88, \quad y_{3,t} = -1.08.$$

- The orthogonal-polynomial kernel fits option prices much better: average pricing-error standard deviation about **\$0.09**, versus about **\$0.63** for the power specification.
- Figure 4 shows that the power kernel is always negatively sloped, but the steepness changes substantially over time.
- Figure 5 shows that the orthogonal-polynomial kernel places more weight on **large negative return states** and sometimes creates a region of locally increasing marginal utility.
- Figure 6 shows that the *average* orthogonal-polynomial kernel resembles earlier findings by Ait-Sahalia and Lo: high valuation of crash states, low valuation of very good states, and a non-monotonic middle region.

Economic message. The data strongly support time variation in the kernel, and the flexible specification captures richer shapes. But richer shape does not automatically mean better economic

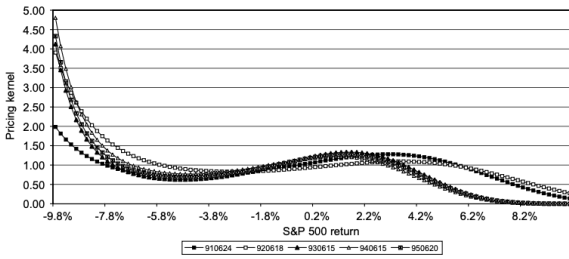


Fig. 5. Empirical pricing kernel using orthogonal polynomial specification in mid-June 1991–1995. We use a four-term expansion: $M_t(r_{t+1}) = \theta_0 T_0(r_{t+1}) + \theta_1 T_1(r_{t+1}) + \theta_2 T_2(r_{t+1}) + \theta_3 T_3(r_{t+1})$. The pricing kernel estimates using this specification assign greater weight to large negative S&P 500 return states and lesser weight to large positive S&P 500 return states than do the estimates using the power specification.

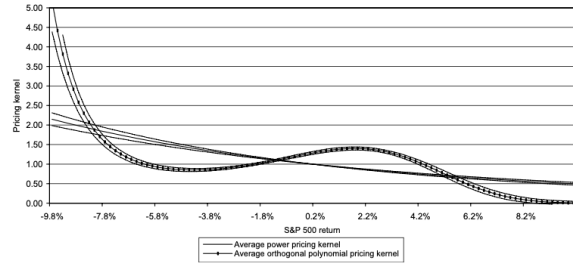


Fig. 6. Time-invariant (average) pricing kernels in mid-June 1991–1995 using the power and orthogonal polynomial specifications. We obtain each average pricing kernel by evaluating the specification at the average parameter estimates for the period from 1991 to 1995. The figure also shows ± 2 standard error bounds for the pricing kernels.

performance.

5.4 Tables 6–7 and Figure 7: empirical risk aversion and the business cycle

Table 6
Summary statistics for empirical risk aversion
We estimate empirical risk aversion over the period 1991–1995, using the exponent of the estimated power pricing kernel. There are seven months for which there is insufficient data for estimation of the EPK, leaving 53 observations of empirical risk aversion.

Number of observations	53
Mean	7.36
Std. dev.	2.58
Min	2.36
Max	12.55
First-order autocorrelation	0.45

Read.

- The power-kernel risk-aversion coefficient has mean **7.36**, standard deviation **2.58**, minimum **2.36**, and maximum **12.55**.
- Its first-order autocorrelation is about **0.45**.
- In the multiple regression, the **credit spread change** enters positively and significantly with coefficient about **9.95**.
- The spread between **implied and objective volatility** also enters positively and significantly.
- In univariate correlations, risk aversion is positively related to the credit spread (**0.50**) and negatively related to the term-structure slope (**−0.36**).
- The regression adjusted R^2 is about **46.1%**.

Economic message. The kernel is not just moving randomly. It moves in a way that looks like **countercyclical risk aversion**: high in bad macro-financial times and low in good times.

5.5 Tables 8–11: hedging as an out-of-sample validation test

Read.

- The hedging sample contains **243 observations** on one-month ATM and OTM S&P 500 puts.

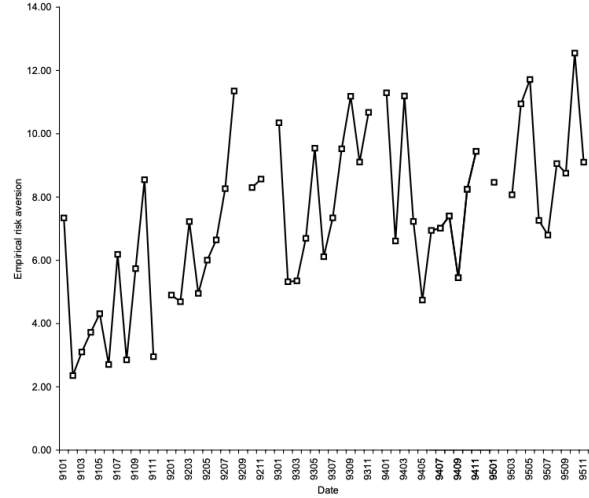


Fig. 7. Empirical risk aversion each month from 1991 through 1995. Empirical risk aversion is the exponent of the estimated power pricing kernel ($\theta_{1,t}$). The seven missing monthly observations are indicated by absence of a data point as well as by a gap in the connecting curve.

Table 8

Summary of option data used for hedging tests

Our hedging sample comprises at-the-money (ATM) and out-of-the-money (OTM) options with approximately one month (from 16 to 24 days) until expiration from the Berkeley Options Database (1991–1995). On each sample date, the ATM put is the closest-to-the-money put (within 1% of the money). The OTM put is the put with moneyness closest to -3% , but no greater than -3% . When we cannot find suitable options with reported prices on the sample date and next trading date, we exclude the sample date from the analysis. We measure the option time-until-expiration in trading days.

	ATM put option	OTM put option
Number of observations	243	243
Average price	\$6.34	\$2.39
Std. dev. price	\$1.04	\$0.77
Average price change	-\$0.11	-\$0.10
Std. dev. price change	\$1.38	\$0.56
Average time to maturity (days)	19.88	19.88
Std. dev. time to maturity (days)	2.40	2.40
Average moneyness ($K/S_t - 1$)	0.00%	-3.61%
Std. dev. moneyness ($K/S_t - 1$)	0.32%	0.48%
Average implied volatility	12.90%	16.36%
Std. dev. of implied volatility	2.79%	2.76%
Minimum implied volatility	9.04%	11.69%
Maximum implied volatility	30.96%	34.39%

Table 9

Time-invariant models of the pricing kernel parameter vector

We estimate the time-invariant power model by averaging the scaling factor ($\theta_{0,t}$) and the risk-aversion coefficient ($\theta_{1,t}$) using the 53 monthly estimates from the EPK estimation. We estimate the time-invariant orthogonal polynomial model by averaging the scaling factor ($\theta_{0,t}$) and generalized Chebyshev coefficients ($\theta_{1,t}, \theta_{2,t}, \theta_{3,t}$) using the 53 monthly estimates from the EPK estimation.

$N = 53$	Estimate
<i>Panel A. Time-invariant power specification</i>	
$\theta_{0,t}$	1.0051
$\theta_{1,t}$	7.3629
<i>Panel B. Time-invariant orthogonal polynomial specification</i>	
$\theta_{0,t}$	0.1945
$\theta_{1,t}$	-2.2457
$\theta_{2,t}$	-0.8791
$\theta_{3,t}$	-1.0752

- The unhedged **\$100** short OTM put position has daily standard deviation about **\$22.56**.
- When hedging only with the underlying index, hedge-portfolio standard deviation falls from **\$12.41** to **\$12.11** when moving from the time-invariant power kernel to the **time-varying EPK power** kernel.
- When hedging with the ATM put, the standard deviation falls from **\$11.21** to **\$11.10** under the power specification.
- When hedging with both the underlying and the ATM put, the power-kernel standard deviation falls from **\$11.36** to **\$11.29**.
- Across the main comparisons, time variation in the pricing kernel improves hedge performance by roughly **1%–3%**.
- The paper also finds that the **power** specification consistently hedges better than the orthogonal-polynomial specification, and most of those pairwise differences are statistically significant.

Table 10

Dynamic models of the pricing kernel parameter vector

We estimate dynamic models of the pricing kernel parameter vector using a linear regression of pricing kernel parameter estimates on one-day lagged parameter estimates. We construct a sample of estimated pricing kernels (power and orthogonal polynomial) on each of the 243 hedging sample dates. We drop from the sample the hedging dates that are not one day apart. We use the remaining 133 observations in the forecasting model estimation regressions. We obtain parameter forecasts each of the 243 hedging dates using the estimated models.

Panel A. EPK power specification

	α	β_0	β_1	Adj- R^2 (%)	Average one-day parameter forecasts	Std. dev. one-day parameter forecasts
$\theta_{0,t+1} = \alpha + \beta_0 \theta_{0,t} + \beta_1 \theta_{1,t} + e_{0,t+1}$						
$\theta_{1,t+1} = \alpha + \beta_1 \theta_{1,t} + e_{1,t+1}$						
$\theta_{0,t+1}$	0.8163	0.1885	0.0000	12.25	1.0053	0.0030
p -val.	<0.0001	<0.0001	0.8632			
$\theta_{1,t+1}$	2.2608		0.6947	49.92	7.4872	1.8717
p -val.	<0.0001		<0.0001			

Panel B. EPK orthogonal polynomial specification

	α	β_0	β_1	β_2	β_3	Adj- R^2 (%)	Average one-day parameter forecasts	Std. dev. one-day parameter forecasts
$\theta_{0,t+1} = \alpha + \beta_0 \theta_{0,t} + \beta_1 \theta_{1,t} + e_{0,t+1}$								
$\theta_{1,t+1} = \alpha + \beta_1 \theta_{1,t} + \beta_2 \theta_{2,t} + \beta_3 \theta_{3,t} + e_{1,t+1}$								
$\theta_{2,t+1} = \alpha + \beta_1 \theta_{1,t} + \beta_2 \theta_{2,t} + \beta_3 \theta_{3,t} + e_{2,t+1}$								
$\theta_{3,t+1} = \alpha + \beta_1 \theta_{1,t} + \beta_2 \theta_{2,t} + \beta_3 \theta_{3,t} + e_{3,t+1}$								
$\theta_{0,t+1}$	0.1756	0.4321	0.0290	-0.0075	-0.0065	39.45	0.2210	0.0698
p -val.	0.0739	0.0996	0.1568	0.8530	0.8961			
$\theta_{1,t+1}$	0.1845		0.8545	-0.4116	0.2034	51.72	-2.0464	0.7858
p -val.	0.0429		<0.0001	0.0556	0.6154			
$\theta_{2,t+1}$	-0.1454		0.1384	0.3496	0.0240	33.60	-0.7196	0.4056
p -val.	0.2665		0.2823	0.0217	0.9330			
$\theta_{3,t+1}$	-0.2535		0.1455	-0.1492	0.5352	45.67	-0.9768	0.3011
p -val.	0.0024		0.0729	0.1166	0.0033			

Table 11

Hedging test results

We conduct hedging tests to compare the (time-varying) EPK power and orthogonal polynomial pricing kernel specifications with the time-invariant (average) power and orthogonal polynomial specifications. The tests cover the period 1991–1995 (243 sample dates). We form hedges of a \$100 position in one-month out-of-the-money (OTM) S&P 500 index put options using one-month at-the-money (ATM) put options, the S&P 500 index portfolio, or both. We construct hedge ratios to neutralize the hedge portfolio sensitivity to the first-order (and in one case, second order) effects of underlying price changes.

Portfolios—pricing kernel specification	Hedge portfolio standard deviation	Reduction in standard deviation (EPK versus time-invariant specification) (%)	Robust t -statistic (EPK versus time-invariant specification)
No hedge			
\$100 OTM written put position	\$22.56		
Hedge using underlying			
Time-invariant power	\$12.41		
EPK power	\$12.11	2.39	1.16
Time-invariant orthog. poly.	\$13.45		
EPK orthogonal polynomial	\$13.13	2.36	1.95
Hedge using ATM put			
Time-invariant power	\$11.21		
EPK power	\$11.10	0.95	2.82
Time-invariant orthog. poly.	\$11.99		
EPK orthogonal polynomial	\$11.64	2.90	1.74
Hedge using underlying and ATM put			
Time-invariant power	\$11.36		
EPK power	\$11.29	0.63	2.94
Time-invariant orthog. poly.	\$12.08		
EPK orthogonal polynomial	\$11.67	3.39	2.12

Economic message. The flexible specification wins on in-sample fit, but the simpler power specification wins on hedging. That is one of the most interesting lessons of the paper.

6 Short summary

- The paper estimates a **date-specific projected pricing kernel** from S&P 500 option prices plus an estimated objective return density.
- The objective density is modeled with an **asymmetric GARCH** process and an empirical innovation density, so both stochastic volatility and fat tails are allowed.
- The pricing kernel is estimated each month as the function that **best fits option prices**.

- Estimated risk aversion varies substantially over time and is **countercyclical**.
- A more flexible orthogonal-polynomial kernel fits option prices better, but a simpler power kernel produces **better hedge ratios**.
- The paper’s main message is that option prices reveal meaningful time variation in state prices and investor risk aversion, but specification choice should be judged by both fit and economic usefulness.

7 Conclusion

7.1 Core conclusion

Rosenberg and Engle show that it is feasible to estimate a *time-varying empirical pricing kernel* from option data once one supplies a dynamic model of the objective return density. The resulting kernel is strongly state dependent, clearly time varying, and informative about the pattern of risk aversion over market states.

7.2 Economic intuition

The key idea is simple. Option prices tell us how much investors are willing to pay for payoffs in different future S&P return states. If we also know how likely those states are, then we can divide “state prices” by “state probabilities” to infer the state-dependent value of one dollar delivered in each state. When the inferred kernel becomes steeper, investors are placing much more value on insurance against bad market outcomes.

7.3 What to remember for class or research

- **Methodologically**, the paper is an important bridge between no-arbitrage option pricing and empirical stochastic-discount-factor estimation.
- **Economically**, it gives evidence that risk aversion is **countercyclical**.
- **Practically**, it shows that a model can fit prices well yet still be weaker for hedging, so model evaluation should not stop at pricing errors.